



**Envision Energy GFM PCS and PPC
PSS/E User Defined Model
User Manual**

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1 Overview

This document provides user necessary information and guidance on using Envision Energy GFM PCS and PPC PSS/E model. It is assumed that the user has a background of GFM PCS control and PSS/E simulation.

The model is developed as a user defined model in PSS/E software to reflect control behavior of Envision Energy GFM PCS and PPC, it allows user to perform RMS dynamic simulation study in PSS/E power system software.

The PCS model receives active power and voltage commands from PPC, or direct settings when there is no PPC. Amounts measurements, control function and protection of the PCS are modelled.

With PPC model implemented, PCS plant active power control, reactive power control and frequency response with different control modes can be realized. In case where Envision PPC is not used in the plant, please make sure Envision PPC model is not included in the .dyr configuration and PCS interface is correctly configured.

However, due to platform limitation, EMT model like PSCAD is a better recommendation for transient study such as fault ride through.

2 Load Flow Model Setup

As the first step in setting up simulations in PSS/E software, the user should prepare a solved load flow case in which the PCS model or PCS aggregation has been properly created.

The PSS/E machine in load flow for each PCS or PCS aggregation should be configured as follows.

- Control mode: Renewable: standard QT, QB limits
- PGEN, QGEN: initial steady state active and reactive power delivered at the machine terminals as required.
- PMAX, PMIN, QMAX, QMIN: the dynamic model does not make use of these values; however, it is good practice to set these according to the capability limits of the Envision PCS unit being modelled. If it is desired to constrain the reactive power to a specific value for the purposes of load flow, please set QMAX and QMIN both equal to QGEN, the target value.
- MBASE: nominal rating of the PCS unit or PCS aggregation. For user's convenience this is based on the nominal active power output: for example, 3.45MW. For a machine representing N identical PCS units, MBASE is N times the MW rating of a single unit. This value is used in dynamic control so it must be set correctly.
- ZR, ZX: these values are not used directly in the dynamic model and do not directly affect behaviour, but their values can influence the numerical performance of the PSS/E simulation engine. The values ZR = 0, ZX = 999pu are recommended.
- RTRAN, XTRAN: must be zero.
- GENTAP: must be 1.

Other parameters may be left as default values.

3 Dynamic Model Configurations

3.1 Dynamic Simulation Setup

With .dll and .dyr files imported, Envision Energy PCS and PPC models are reflected as:

- **ENGFMxx:** represents the Envision GFM PCS controller model and is written as a 'renewable model'. It receives commands from PPC or interface settings. Amounts measurement, steady state operation, transient state control and protection have been modelled.
- **BNPPC_GFMVx:** represents the power plant controller (PPC) model and is written as an 'other model'. It issues active power and voltage commands to individual PCS units or aggregates in order to regulate active power, frequency and/or reactive power or voltage at the plant point of connection.

The PSS/E software uses a modified load flow engine to calculate quasi-static network solutions during a dynamic simulation. This engine generally performs well but is sometimes limited by its ability to solve networks with low short-circuit ratios and high transient power flows.

When using the Envision dynamic model with generators whose power output is large compared with the short-circuit fault level at the point of connection, slight modification of PSS/E's solution parameters may assist with obtaining reliable network solutions when simulating the response to solid three-phase faults close to the point of connection. Suggested modifications include reduction of the 'acceleration factor' to 0.5 or 0.3 from its default value of 1.0, and an increase in the maximum iteration count to 100 or 200 from its default value of 25.

With certain networks, similar modifications to the solution parameters may be necessary when simulating system response to severe faults, or to faults remote from the PCS terminals.

3.2 PCS Model Configuration

3.2.1 BESS System Rating

Parameters in the table below define the rating and overload of the BESS system based on generator Mbase.

3.2.2 Command Interface and Priority

The PCS model can receive commands from either PPC or direct interfaces, via settings ICON(M+3). Details on the configurable parameter and command interface are given in Table 2 below.

Table 1 PCS model PQ control configurable settings

ICON(M+3)	Command Sent From	Command Interface	Unit
0	PPC	'WPCMND' and 'VREF'	p.u. (power on generator Mbase)
1	Local control	Var(L) and V(L+2)	p.u.

3.2.3 Voltage and Reactive Power Control

The PCS controls reactive power by adjusting its own output voltage magnitude, thereby supporting the voltage at the point of connection. The PCS controls via Q-V droop control: when the voltage is low, the PCS increases reactive power output in response; when voltage becomes high, the reactive power is adjusted according to the droop characteristic.

3.2.4 Active Power Control

The PCS controls active power by adjusting its output frequency and phase angle, thereby providing frequency support.

3.2.5 Primary Frequency Control

In the model provided by default, the primary frequency control function is disabled. With the local primary frequency control function enabled, incremental active power command is calculated using a frequency control droop coefficient based on PCS terminal frequency and base frequency value; the incremental active power command is then added into the active power control loop as described in Section 3.2.3.

Relevant control parameters for primary frequency control are as shown in Table 2 below.

Table 2 Primary Frequency Control Settings

ICON/CON	CON Name	Description	Unit
CON(J+12)	pKdroop	Droop coefficient for primary f control	N/A
CON(J+30)	Wdb+	Positive deadband for primary f control	p.u.
CON(J+31)	Wdb-	Negative deadband for primary f control	p.u.

3.2.6 Protection

With ICON(M+6) set as 0, protection function is enabled. Voltage and frequency protections are implemented, and user could adjust CON(J+65) to CON(J+100) for low frequency protection, high frequency protection, low voltage protection and high voltage protection. The protection function operates on a conventional ‘threshold and timer’ basis; when either of the voltage/frequency level and time duration sets is met, the protection will be triggered and it will trip the PCS model.

In case for testing purpose, setting ICON(M+6) being 1 can disable the protection function.

It needs to be noticed that the extreme low frequency protection is only valid for discharging model; the extreme high frequency protection is only valid for charging model. Please refer to PCS product specification for details.

3.3 PPC Model Configuration

3.3.1 PPC Model General Settings

The Envision PPC model is written as a Bus Type Model, which can be attached to any existing bus; its variable index can be get by `psspy.cctmind_buso( )`.

For different controlled models, the dynamic configuration must be changed accordingly, specifically as below:

- **Power Flow and Controlled Bus:** the power flow from bus ICON(M+8) to bus ICON(M+9), and the voltage at bus ICON(M+6) will be controlled for voltage control mode, and frequency at bus ICON(M+7) will be controlled for frequency response.
- **Setting for Controlling Multiple Generators:** Envision PPC model can support up to 300 generators (or aggregates, one generator could be one aggregate, same below) in total, as in up to 6 groups. ICOM(M+21) to ICON(M+38) defines the settings of 6 groups separately. Settings of the first group is introduced below and the same way for the other groups.
 - When there is only 1 generator in each group, set ICON(M+39) as 0 and ICON(M+22) as ID of the generator. Alternatively, the setting approach for more groups can also be adopted if all the generators share the same ID.
 - When there are more than 1 generators in one of the groups, the ID of all generators must be the same and the generator attached bus number in each group must be in sequence (eg: 71000, 71001....). Set ICON(M+39) as 1, ID of the generators is set in ICON(M+20). For Group 1 for example, the starting generator bus number is set by ICON(M+21) and ICON(M+22) is number of the generators in the group.
- **Generator Type Setting:** Different types of generators can be controlled by the PPC. However, in each group, the generators must share the same type and reactive power command limitation. ICON(M+23) defines type of the generator being controlled in the first group for example, setting description can be found in the table below. PPC command limitation is in reactive power format in p.u. based on Mbase setting unless otherwise stated. And same for the other groups. Table 6 below describes the setting for ICON(M+23).

Table 3 ICON(M+23) generator type setting description

ICON(M+23)	Description
0	WTG is controlled and PPC command limitation is in power factor Cos value format
1	WTG is controlled and PPC command limitation is in reactive power format in p.u. based on Mbase
2	BESS PCS is controlled
3	Machine type STATCOM is controlled
4	FACTS type STATCOM is controlled; and the STATCOM command input VAR index number should be set by ICON(M+40)

3.3.2 Active Power Control and Frequency Response

In regards to active power and frequency control, the PPC model can provide AGC active power control, inertia response and frequency control of different control modes. Control mode can be defined via ICON(M+1) as described in Table 7. Control logics of each function and each control mode are detailed below.

Table 4 ICOM(M+1) Active power control mode selection

ICOM(M+1)	Control mode
0	No P control or f response

ICOM(M+1)	Control mode
1	P control AGC function enabled
2	Primary frequency response function is enabled, with AGC enabled; active power regulation result is the summation of AGC and frequency response

Under AGC active power control, the plant active power reference can be given via interface Var(L+3), in per unit on generator Mbase.

Under frequency control, the plant active power regulation amount calculation follows the formula below:

$$droop(\%) = 100 \times \frac{\Delta f / f_{nom}}{\Delta P / P_n}$$

Where:

Δf is the measured frequency deviation beyond deadband, in Hz;

ΔP is the active power change amount, in MW;

f_{nom} is the nominal frequency 50 or 60, in Hz;

P_n is the controlled rated power of the PPC, in MW;

droop is the adjustable droop setting and can be adjusted via CON(J+15), in %.

Deadband can be set via CON(J+32) and CON(J+33), limitation on active power regulation amount can be adjusted via CON(J+40) and CON(J+41).

A few other functions and settings for active power and frequency control are detailed below:

Active power limit: Plant active power output can be limited via CON(J+30).

PPC freezing when frequency deviates: When POC frequency deviation gets above CON(J+82), the PPC will freeze active power command sent to the PCS units, until frequency recovers. When POC frequency deviation gets higher than CON(J+83), the PPC P control PI controller output will stop increasing if the frequency deviation is positive or stop decreasing when the frequency deviation is negative.

3.3.3 Reactive Power/Voltage Control

In regards to reactive power control, the PPC model can provide four different control modes that can be defined via ICOM(M) as described in Table 8 below.

Table 5 ICOM(M) Reactive power control mode selection

ICOM(M+1)	Control mode	Setpoint interface
0	PPC Q control is disabled	/
1	Reactive power control	Qref input from Var(L) in p.u. on sum of Mbase of all active generators and Qmax of STATCOMs (if there is any)
2	power factor control mode	PFref input from Var(L+1) in cos value
3	voltage droop control mode	Vref input from Var(L+2) in p.u.
4	direct voltage control mode	Vref input from Var(L+2) in p.u.

- Reactive power control mode and power factor control mode

In reactive power control mode, the PPC targets to regulate control point Q to setpoint value via a PI control loop; in power factor control mode, the PPC calculates Qref from power factor setpoint and plant active power and uses the Q control loop to achieve Qref. In these two modes, PI parameters can be adjusted via CON(J+24) and CON(J+25).

- Voltage droop control mode

In voltage droop control mode, PPC calculates Qref as per the droop formula below, and Qref goes into a PI control loop, the PPC targets to regulate control point Q to the dynamically calculated Qref value.

$$Q_{ref_MVAr} = droop \times (V_{ref_pu} - V_{meas_pu}) \times Base$$

Where:

V_{ref_pu} is the voltage setpoint received via Var(L+2), in p.u.;

V_{meas_pu} is the measured voltage at control point, in p.u.;

droop is the adjustable droop setting and can be adjusted via CON(J+16) and CON(J+17);

Base is an adjustable setting and can be set via CON(J+29), if it is set as 0 then the controlled generator Mbase will be used.

Voltage control deadband can be set via CON(J+20) and CON(J+21), PI parameters can be set via CON(J+22) and CON(J+23).

- Direct voltage control mode

In direct voltage control mode, PPC targets to regulate control point V to setpoint value via a PI control loop. Voltage control deadband can be set via CON(J+20) and CON(J+21), PI parameters can be set via CON(J+68) and CON(J+69).

A few other functions and settings for reactive power are detailed below

Reactive power limitation: Generator Q command limitation can be set in CON(J+50) to CON(J+61) in p.u. for each generator group. The plant reactive power capability is limited by CON(J+18) and CON(J+19) in p.u.

PPC freezing when voltage deviates: When POC voltage gets lower than CON(J+11) or higher than CON(J+12), the PPC will freeze active power and voltage commands sent to the PCS units, until voltage recovers. When POC voltage gets higher than CON(J+64), the PPC Q command in Q control mode or the value calculated in power factor control mode will stop increasing; when POC voltage gets lower than CON(J+65), it will stop decreasing.